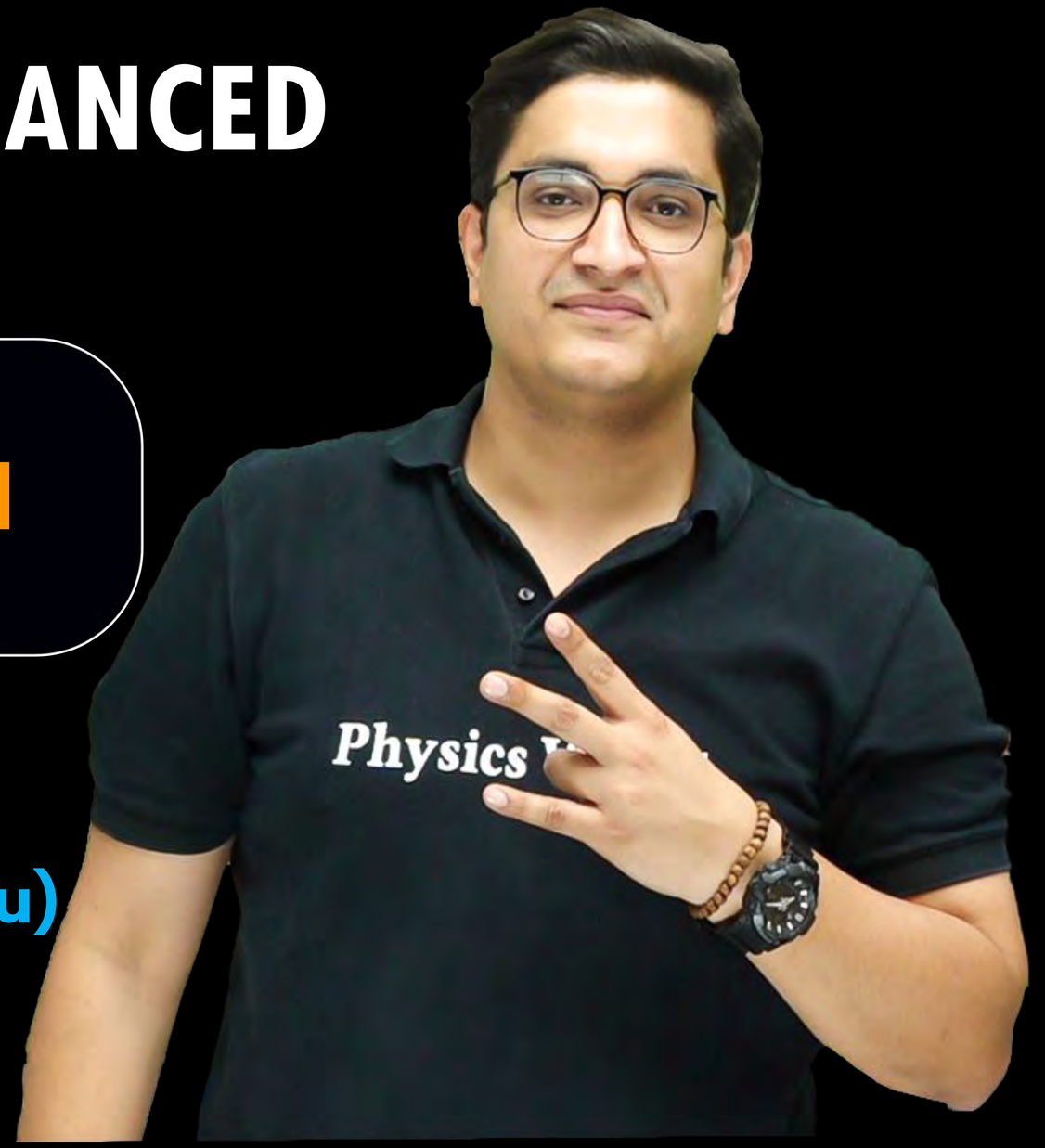
Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

QUADRATIC EQUATION



DIBY (DO IT BY YOURSELF)- A to M

By: Sachin Jakhar (Defender Mamu)

CHALLENGER QUESTIONS

Acidic Question

If the equations $x^3 + x^2 - 4x = 4$ and $x^2 + px + 2p = 0$ $(p \in R)$ have two roots common, then the value of p is

DIBY-A

A -2

B -1

C 1

D 3

Acidic Question

The values of k, for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ posses atleast one positive root, are

DIBY-B

A $[4, \infty)$

B $(-\infty, -1) \cup [4, \infty)$

C $[-1, 4]$

D $(-\infty, -1]$

Challenger Question

If the roots of the equation $x^2 + (p+1)x + 2q - q^2 + 3 = 0$ are real and unequal $\forall\ p \in R$, then find the minimum integral value of $(q^2 - 2q)$.

DIBY-C

Challenger Question [JEE Mains-2021]

If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to

DIBY-D

A 56×3^{25}

B 56×3^{24}

C 52×3^{24}

D 25×3^{25}

Question

If α, β and γ are roots of $3x^3 - 4x^2 - 3x + 2 = 0$ and

$$(\alpha^5 + \beta^5 + \gamma^5) - (\alpha^3 + \beta^3 + \gamma^3) = \frac{2}{m}\left(2(\alpha^4 + \beta^4 + \gamma^4) - (\alpha^2 + \beta^2 + \gamma^2)\right)$$

Then value of m is

DIBY-E

Question

Let α, β, γ be roots of $x^3 + 2x^2 - 4x + 5 = 0$ find value of $\frac{(\alpha^3+5)(\beta^3+5)(\gamma^3+5)}{13\alpha\beta\gamma}$.

DIBY-F

Challenger Question [JEE Mains-2024]

Let $x_1, x_2, x_3, x_4,$ be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_3^2\right)\left(4 + x_4^2\right) = \frac{125}{16}m.$ Then the value of m is _____.

DIBY-G

Critical Thinking Question (CTQ)

Let α, β, γ be distinct real number and $f(x)$ is a quadratic polynomial such that

$f(2)\alpha^2 + f(3)\alpha + f(4) = 4\alpha^2 + 4\alpha + 8$

$f(2)\beta^2 + f(3)\beta + f(4) = 4\beta^2 + 4\beta + 8$

$f(2)\gamma^2 + f(3)\gamma + f(4) = 4\gamma^2 + 4\gamma + 8$

then find the value of $f(7)$.

DIBY-H

Question [JEE Mains-2023]

Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$.
If $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to

A 155/8

B 21

C 169/8

D 19

DIBY-I

Challenger Question [Multiple Correct]

DIBY-J

If $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are roots of equation $x^5 - 5x^4 - 1 = 0$, then

A $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -\frac{1}{20}$

B $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -20$

C $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^5 = 1$

D $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^3 = \frac{1}{5}$

Acidic Question

If α, β are roots of $ax^2 + bx + c = 0$, then the equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ has roots

DIBY-K

A $\dfrac{\alpha}{1-\alpha}, \dfrac{\beta}{1-\beta}$

B $\dfrac{1-\alpha}{\alpha}, \dfrac{1-\beta}{\beta}$

C $\dfrac{\alpha}{1+\alpha}, \dfrac{\beta}{1+\beta}$

D $\dfrac{1+\alpha}{\alpha}, \dfrac{1+\beta}{\beta}$

Question [JEE Mains-2021]

If $x^2 + 9y^2 - 4x + 3 = 0, x, y \in R$, then x and y respectively lie in the intervals.

DIBY-L

A $[1, 3]$ and $[1, 3]$

B $[1, 3]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

C $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $[1, 3]$

D $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

Critical Thinking Question (CTQ)

DIBY-M

a, b, c, d are real numbers such that

$(a+1)(b+1)(c+1)(d+1) = 1$

$(a+2)(b+2)(c+2)(d+2) = 2$

$(a+3)(b+3)(c+3)(d+3) = 3$

$(a+4)(b+4)(c+4)(d+4) = 4$

Find $(a+5)(b+5)(c+5)(d+5)$.

ANSWERS (A TO M)

DIBY-A : B

DIBY-B : D

DIBY-C : 4

DIBY-D : C

DIBY-E : 3

DIBY-F : 8

DIBY-G : 221

DIBY-H : 44

DIBY-I : D

DIBY-J : B,C

DIBY-K : C

DIBY-L : B

DIBY-M : 29

It is never about end result

It's always about Journey

THANK You

#futureIITians